Tangents and normals

1

a

i $y' = \frac{1}{x}$ ii $\frac{1}{5}$

b

i $y' = 4 + \frac{1}{x}$ ii 9

c

i $y' = \frac{2x}{x^2 + 5}$ ii $\frac{3}{7}$

d

i $y' = \frac{4x^3}{x^4 + 4}$ ii $\frac{8}{5}$

e

i $y' = 4x \ln x + 2x$ ii $6e$

2

a

i $-\frac{2}{5}$ ii $y = -\frac{2x}{5} + \ln 2 + \frac{2}{5}$

b

i -5 ii $y = -5x + \ln 5 + 25$

c

i $-3 \ln 3$ ii $y = -3x \ln 3 + 9 \ln 3 + 1$

3

a $x = 1$ b $y = x - 1$ c $y = 1 - x$

4

a $x < 0$ b $\frac{dy}{dx} = \frac{1}{x}$ c $-\frac{1}{5}$

5

a $y' = \frac{4x}{x^2 + e}$ b 0 c $y = 2$

6

a $y' = \frac{1}{x \ln x}$ b 0 c $y = \frac{1}{e}x - 1$

7

a $y' = \frac{9x^2}{x^3 + 2}$ b 3

c $y = -\frac{x}{3} + 3 \ln 3 + \frac{1}{3}$

8

a $y' = \frac{-4}{\ldots} + 2$ b $y = -0.97$ c $x = 2$

9 $a = -3$

Graphs of logarithmic functions

10

a

i $x < 0$

ii $\frac{1}{x}$

iii $x < 0$

b

i $x > 7$

ii $\frac{1}{x-7} + \frac{1}{x}$

iii $x > 7$

11

a Increasing

b No

c It decreases at a decreasing rate.

d It increases towards ∞.

12

a $y = -3\ln(x-4)$

b $\frac{dy}{dx} = -\frac{3}{x-4}$

c Undefined when $x \leq 4$

d Strictly decreasing as the derivative is always positive if $x > 4$

13

a $f'(x) = \frac{4}{x}$

b 2

c $f''(x) = -\frac{4}{x^2}$

d -1

e Increasing

f Concave down

14

a $x > 0$

b $\left(\frac{1}{\sqrt{e}}, -\frac{1}{2e}\right)$

c Minimum

15

a $x = e$

b Maximum

16

a $x > 0$

b $y' = 1 + \ln x$

c $\left(e^{-1}, -e^{-1}\right)$

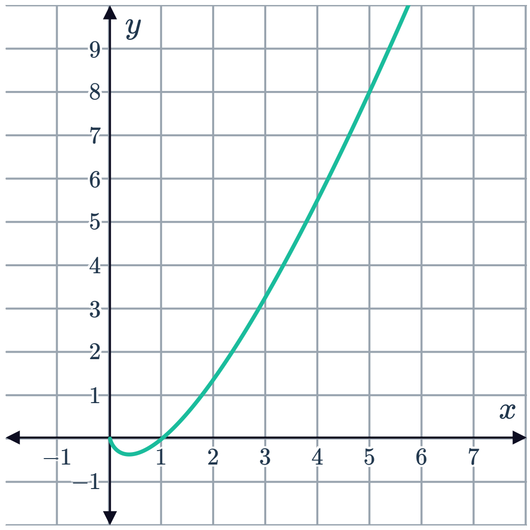
d $y'' = e$

e Minimum

f $y \geq -\dfrac{1}{e}$

g As $x \to \infty, y \to \infty$.

h



17

a $y' = 1 - \dfrac{1}{x}$

b $(1, 1)$

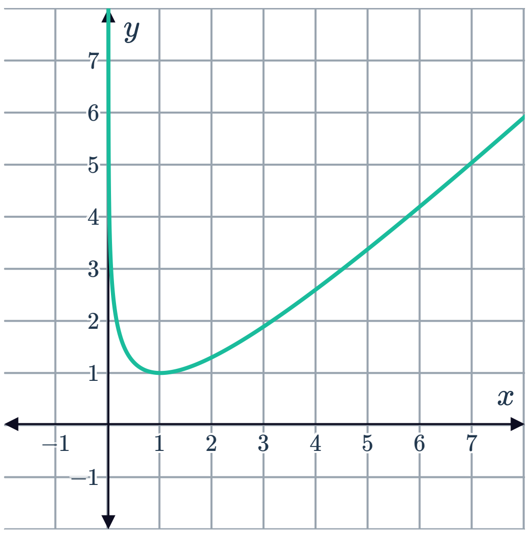
c $y'' = \dfrac{1}{x^2}$

d Minimum

e As $x \to \infty, y \to \infty$.

f As $x \to 0, y \to \infty$.

g



18

a $x = e$

b $(e, 0)$

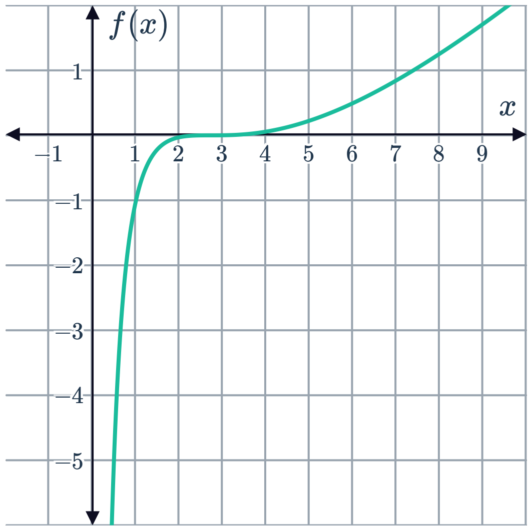
c $f''(x) = \dfrac{3(\ln x - 1)(3 - \ln x)}{x^2}$

d

x	1	e	e^3	e^4
$f''(x)$	-9	0	0	$-9e^{-8}$

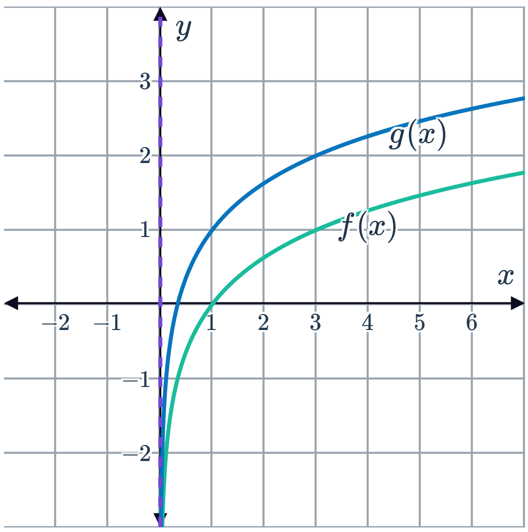
e $(e, 0), (e^3, 8)$

f



19

a



b $f'(x) = \dfrac{1}{x}$

c $g'(x) = \dfrac{1}{x}$

d The tangents to the curve are parallel and increasing at the same rate for any $x > 0$.

Applications

20

a $\frac{dh}{dt} = \frac{600}{t+1}$

b 120 m/min

c Decreasing rate

21

a $m = 3400$. It is the number of passengers serviced in the initial month when $(t = 0)$.

b $k = 150$

c $t = 9$

d $P'(0) = 150$

e $t = 1$ month

22

a $t \geq 0$

b $\frac{136}{101}$

c $t = 10$

d

i $A'(10) = \frac{8}{5}$

ii $B'(10) = \frac{6}{5}$

e Species A

23

a $A = 10.1$

b $k = -4.5$

c $P'(4) = -0.9$

d $t = 9$ years

e $P'(9) = -0.45$